

Improving Adversarial Robustness in Face Anti-Spoofing via Transfer Learning Strategies and Dynamic Adversarial Training on Swin Transformer

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Abstract—Face recognition systems deployed in critical environments are exposed to two complementary types of threats: presentation attacks (spoofing) and adversarial attacks based on imperceptible mathematical noise. In this work, we evaluate the adversarial vulnerability of a Swin Transformer (Tiny version), pre-trained on ImageNet and fine-tuned for spoof detection on the LCC-FASD dataset, against FGSM and PGD attacks. We compare five transfer learning strategies (*Freeze All*, *1-Fine-tune*, *2-Fine-tune*, *3-Fine-tune*, *Fine-tune All*) to identify the most robust configuration. Phase 1 results show that partial unfreezing of two stages (*2-Fine-tune*) achieves the best trade-off with 98.18 % nominal accuracy and 60.05 % under PGD attack. In Phase 2, we apply dynamic on-the-fly adversarial training (PGD, $\epsilon = 2/255$, 10 iterations, 50/50 ratio) on this model, raising PGD robustness to 86.74 % (+26.69 points) while maintaining clean accuracy at 98.48 %. These findings provide practical guidelines for deploying adversarially robust facial biometric systems.

Index Terms—Face Anti-Spoofing, Swin Transformer, Transfer Learning, Adversarial Attacks, FGSM, PGD, Adversarial Training, LCC-FASD, Vision Transformer.

I. INTRODUCTION

Face biometrics has become one of the most widely deployed authentication modalities, integrated into applications ranging from smartphone unlocking to industrial access control systems [1]. Its ease of use, combined with advances in Deep Learning architectures, provides remarkable accuracy under controlled conditions.

However, this popularity attracts increasingly sophisticated adversaries. On one hand, presentation attacks (*spoofing*) exploit physical artifacts — 3D masks, printed photos, replayed videos on screens — to fool the sensor [2]. On the other hand, adversarial attacks add imperceptible mathematical perturbations that drastically degrade the performance of a neural network [3]. Vision Transformers, and in particular the Swin Transformer [4], represent the state of the art in Face Anti-Spoofing (FAS); yet their robustness against adversarial noise remains poorly studied.

A high-performing FAS model under clean conditions can therefore collapse under a targeted adversarial attack. The central question is: *does the transfer learning strategy used to specialize the Swin Transformer significantly influence its adversarial robustness?*

This work pursues two complementary objectives. In **Phase 1**, we measure and compare the natural robustness of

five fine-tuning strategies under FGSM and PGD attacks. In **Phase 2**, we apply dynamic adversarial training on the optimal strategy identified, in order to push its robustness limits further. This paper is organized as follows: Section II presents materials and methods, Section III details the results of both phases, Section IV discusses the findings, and Section V concludes with future perspectives.

II. MATERIALS AND METHODS

A. Dataset and Pre-processing

The LCC-FASD dataset (*Large Corpus for Cross-database Face Anti-Spoofing Detection*) is used to train and evaluate our models. It contains **18,827 images** distributed across two highly imbalanced classes: 16,885 spoof images and only 1,942 real images, yielding a ratio of $\approx 8.7:1$ (Figure 1).

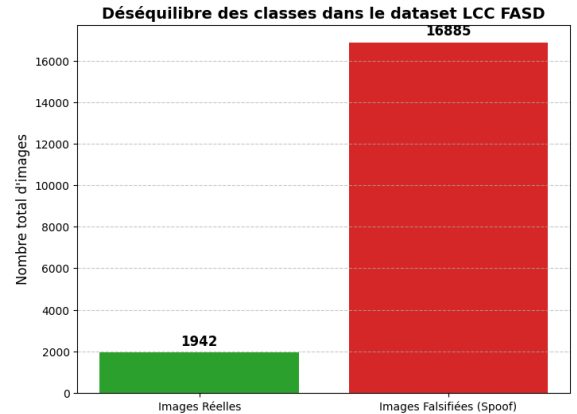


Fig. 1. Class imbalance in LCC-FASD: 16,885 spoof images vs. 1,942 real images (ratio $\approx 8.7:1$). The severe imbalance motivates the use of `WeightedRandomSampler` during training.

This imbalance can introduce severe learning bias. To address this, we implement a PyTorch `WeightedRandomSampler` that ensures balanced class representation in each mini-batch, without naive over-sampling. Figure 2 shows examples of real faces and presentation attacks from the dataset.

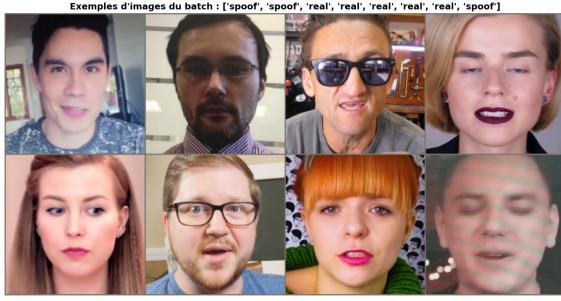


Fig. 2. Sample batch from LCC-FASD showing a mix of real faces and spoof images (printed photos, screen replays). The label bar reads: ['spoof', 'spoof', 'real', 'real', 'real', 'real', 'real', 'spoof'].

Images are resized to 224×224 pixels and normalized using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

B. Architecture: Swin Transformer

The Swin Transformer [4] is a hierarchical Vision Transformer that computes self-attention within shifted local windows (W -MSA and SW -MSA), reducing the quadratic complexity of standard Transformers to linear complexity. Its *Tiny* version processes images as a four-stage pyramid (Figure 3), where each stage halves the spatial resolution and doubles the channel depth via a learned *Patch Merging* operation.

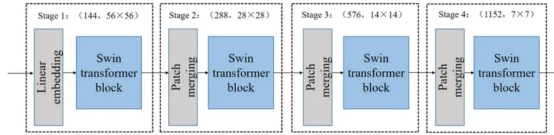


Fig. 3. The 4-stage hierarchical architecture of the Swin Transformer Tiny. Each stage reduces spatial resolution while increasing feature dimensionality: Stage 1 (56×56), Stage 2 (28×28), Stage 3 (14×14), Stage 4 (7×7). The 2-Fine-tune strategy unfreezes Stages 3–4 and the classification head.

The four stages are described below, and their main characteristics are summarized in Table I.

Stage 1 – Low-level Feature Extraction (56×56). The input image ($224 \times 224 \times 3$) is split into non-overlapping 4×4 patches, each projected into a 96-dimensional embedding, yielding a feature map of size $56 \times 56 \times 96$. Two Swin Transformer blocks (W -MSA then SW -MSA) capture basic visual patterns — edges, corners, color gradients, and low-level textures — in a purely local fashion. These generic features closely resemble what early CNN layers detect.

Stage 2 – Mid-level Features (28×28). Patch Merging groups 2×2 neighboring patches and projects them, halving spatial resolution to $28 \times 28 \times 192$. Two Swin blocks learn composite patterns such as shapes, object parts, and local spatial structures. For FAS, Stage 2 begins to encode texture differences between real skin and printed paper or screen pixels.

Stage 3 – High-level Semantic Features (14×14). Another Patch Merging yields $14 \times 14 \times 384$. With **six** Swin Transformer blocks, Stage 3 is computationally the richest stage. It captures

high-level semantic representations including face parts (eyes, nose, mouth), moiré patterns, screen borders, and mask surface artifacts — the features most discriminative for FAS. *This is precisely why unfreezing Stage 3 is the pivotal decision in the 2-Fine-tune strategy.*

Stage 4 – Global Representation (7×7). A final Patch Merging gives $7 \times 7 \times 768$. Two Swin blocks aggregate all local cues into a compact, holistic face representation. An Adaptive Average Pooling then collapses the 7×7 map into a single 768-dimensional vector, which is fed into the binary classification head (Linear $\rightarrow$ Real/Spoof).

TABLE I
SWIN TRANSFORMER TINY – PER-STAGE ARCHITECTURE SUMMARY.

Stage	Resolution	Channels	Blocks	Features learned
1	56×56	96	2	Edges, textures, colors
2	28×28	192	2	Shapes, local patterns
3	14×14	384	6	Face parts, spoof artifacts
4	7×7	768	2	Global face representation

C. Transfer Learning Strategies

We start from an ImageNet pre-trained Swin Transformer Tiny and replace the final classification head with a two-output linear layer (Real / Spoof). The key design choice is *how many backbone stages to unfreeze* during fine-tuning on LCC-FASD. More unfrozen stages means more trainable parameters and higher task-specific accuracy, but also greater risk of destroying the generalizable ImageNet representations that provide natural adversarial resilience. Five strategies are evaluated:

Freeze All. Every backbone layer is frozen; only the head is trained. The backbone acts as a fixed ImageNet feature extractor. The model achieves decent accuracy (94.47 %) but collapses under iterative PGD attack (5.22 %) because no layer was ever optimized for FAS-specific cues.

1-Fine-tune. Stage 4 and the head are unfrozen. Stage 4 re-learns to re-interpret the global representation for FAS while Stages 1–3 remain frozen. This yields a moderate improvement in both accuracy (96.56 %) and PGD robustness (24.05 %) compared to Freeze All.

2-Fine-tune (optimal). Stages 3–4 and the head are unfrozen. Unlocking Stage 3 — the richest stage with six blocks — allows the model to re-learn high-level FAS discriminative features (moiré patterns, paper grain, screen glow) while Stages 1–2 remain frozen and preserve the robust low-level ImageNet representations. This partial inheritance yields the best trade-off: 98.18 % clean accuracy and 60.05 % robustness under PGD.

3-Fine-tune. Stages 2–4 and the head are unfrozen. Unlocking Stage 2 starts to modify mid-level feature detectors. Although clean accuracy slightly improves (98.72 %), PGD robustness drops to 37.32 %, showing that modifying Stage 2 begins to erode the stable generalizable representations.

Fine-tune All. All layers including Stage 1 are retrained on LCC-FASD. The model entirely forgets ImageNet knowledge

and overfits to the exact distribution of LCC-FASD. Clean accuracy remains high (97.74 %) but PGD robustness collapses to **0.00 %** — a textbook example of *adversarial overfitting*.

D. Adversarial Threat Modeling

Adversarial attacks exploit the fact that neural network loss landscapes are not smooth: there exist directions in pixel space where a tiny step causes a large jump in the predicted class, while the image remains visually unchanged to a human observer. The attacker computes these directions by back-propagating the loss with respect to the *input pixels* (not the weights).

1) *FGSM – Fast Gradient Sign Method*: Proposed by Goodfellow et al. [3], FGSM perturbs every pixel by exactly ε in the direction that maximally increases the classification loss, in a single gradient step:

$$x^* = x + \varepsilon \cdot \text{sign}(\nabla_x \mathcal{L}(\theta, x, y)) \quad (1)$$

where x is the clean image, y its true label, θ the model parameters, and ε the L_∞ perturbation budget.

The attack proceeds as follows: (1) perform a forward pass to compute the loss $\mathcal{L}$; (2) back-propagate to obtain $\nabla_x \mathcal{L}$; (3) retain only the sign (+1 or -1) of each gradient component — the direction of steepest loss increase; (4) scale by ε and add to the original image. Even a change of $\varepsilon = 0.01$ per pixel — invisible to the eye — accumulates enough signal across thousands of pixels to push the model’s prediction across the decision boundary.

FGSM is a **one-shot attack**: it takes a single step in the gradient direction and is therefore fast but not always optimal, since the gradient direction at x may not be the globally best attack direction.

2) *PGD – Projected Gradient Descent*: Proposed by Madry et al. [5], PGD refines the FGSM idea by iterating the gradient step N times with a smaller step size α , projecting back onto the $L_\infty(\varepsilon)$ ball after each step to enforce the perturbation budget:

$$x^{t+1} = \Pi_{B_\varepsilon(x)}(x^t + \alpha \cdot \text{sign}(\nabla_{x^t} \mathcal{L}(\theta, x^t, y))) \quad (2)$$

The attack proceeds as follows: (1) initialize $x^0 = x$ (or add small random noise for a stronger variant); (2) repeat N times: compute the gradient at the current perturbed image, take a step of size α in the sign direction, then clip each pixel so that $|x^t - x| \leq \varepsilon$; (3) return the final x^N as the adversarial example. By exploring the loss landscape iteratively, PGD reliably finds a near-optimal adversarial example within the ε -ball. Madry et al. proved that PGD is essentially the **strongest first-order attack** possible, making it the standard benchmark for adversarial robustness evaluation.

Figure 4 illustrates the effect of PGD on face images.

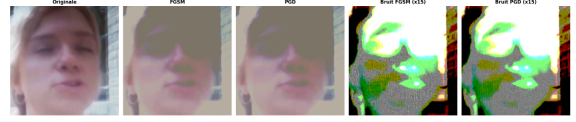


Fig. 4. Effect of PGD adversarial perturbation on face images. From left to right: original clean image, perturbed image ($\varepsilon = 0.01$) visually identical to the clean one, and the amplified perturbation noise ($\times 10$) revealing the adversarial pattern. The rightmost columns show the attack applied on a spoof sample.

Table II compares the two attacks and lists the hyperparameters used in each experimental phase.

TABLE II
FGSM VS. PGD – COMPARISON AND HYPERPARAMETERS USED.

Property	FGSM	PGD
Steps (N)	1	10
Step size (α)	ε	0.5/255
Attack strength	Moderate	Strong
Computation cost	Very fast	$\approx N \times$ slower
ε Phase 1	0.01	0.01
ε Phase 2	–	8/255

E. Defense Strategy: Dynamic Adversarial Training

In Phase 2, we implement on-the-fly adversarial training on the best strategy identified in Phase 1 (*2-Fine-tune*). At each training iteration, 50 % of batch images are replaced by their adversarial versions generated by PGD in real time. By regenerating adversarial examples at every batch, the model is exposed to a continuously varying set of perturbations and is forced to learn representations that are genuinely invariant to adversarial noise, rather than overfitting to a fixed set of pre-computed attacks. The hyperparameters are: $\varepsilon = 8/255$, $\alpha = 0.5/255$, $N_{\text{steps}} = 10$, adversarial ratio = 0.5, $\text{lr} = 5 \times 10^{-5}$, 10 epochs. The low learning rate preserves the convergence achieved in Phase 1 while allowing the model to adapt its decision boundaries to adversarial directions.

III. RESULTS

A. Phase 1: Transfer Strategy Benchmark

Table III reports the global accuracy of the five strategies under clean conditions, under FGSM attack, and under PGD attack ($\varepsilon = 0.01$). Figure 5 presents these results as grouped bar charts.

TABLE III
ACCURACY (%) OF THE SWIN TRANSFORMER UNDER DIFFERENT
TRANSFER LEARNING STRATEGIES AND ADVERSARIAL ATTACKS
($\varepsilon = 0.01$).

Strategy	Clean	FGSM	PGD
Freeze All	94.47 %	67.72 %	5.22 %
1-Fine-tune	96.56 %	86.81 %	24.05 %
2-Fine-tune	98.18 %	93.79 %	60.05 %
3-Fine-tune	98.72 %	92.65 %	37.32 %
Fine-tune All	97.74 %	36.41 %	0.00 %

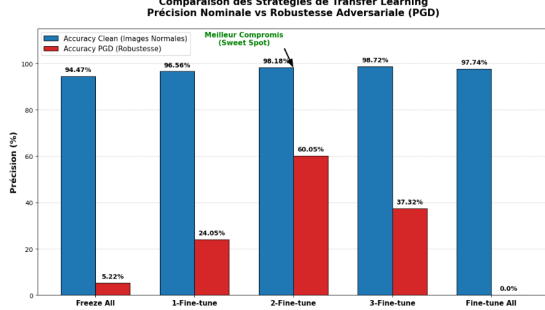


Fig. 5. Comparison of the 5 transfer learning strategies: nominal accuracy (clean) vs. adversarial robustness under PGD. The 2-Fine-tune bar achieves the highest PGD robustness (60.05 %) while maintaining top-tier clean accuracy.

The results reveal a clear trend: as unfreezing increases, nominal accuracy improves up to a threshold, but adversarial robustness collapses. The extreme case *Fine-tune All* reaches 0.00 % under PGD — a catastrophic degradation. Conversely, complete freezing *Freeze All* retains few FAS-specific features and drops to 5.22 % under PGD. The 2-Fine-tune strategy stands out with its optimal compromise.

B. Phase 2: Dynamic Adversarial Training

Starting from the 2-Fine-tune weights saved in Phase 1, we apply on-the-fly adversarial training for 10 epochs with PGD-generated examples ($\varepsilon = 8/255$, 10 steps, 50 % adversarial ratio, $\text{lr} = 5 \times 10^{-5}$). Table IV reports the full per-epoch training log. Two observations stand out. First, the training loss decreases monotonically from 0.4096 to 0.1999, confirming stable convergence throughout the 10 epochs. Second, the training accuracy climbs steadily from 72.8 % at Epoch 1 to 90.7 % at Epoch 10 — a direct measure of how well the model learns to correctly classify the adversarially perturbed images it encounters during training. Validation accuracy on clean data remains consistently high (between 96.9 % and 98.8 %) across all epochs, demonstrating that adversarial training does not degrade performance on clean images. The best validation checkpoint (marked †) was recorded at Epoch 1 with 98.8 % validation accuracy and an F1-score of 97.45 %.

Figure 6 shows the training loss and validation accuracy curves across the 10 epochs.

TABLE IV
PER-EPOCH TRAINING LOG FOR PHASE 2 ADVERSARIAL TRAINING
($\varepsilon = 8/255$, PGD STEPS= 10, RATIO=0.5). † MARKS THE BEST
VALIDATION CHECKPOINT (SAVED MODEL).

Epoch	Loss	Train Acc.	Val Acc.	F1 Score
01†	0.4096	72.8 %	98.8 %	97.45 %
02	0.3564	74.9 %	97.4 %	94.66 %
03	0.3437	78.2 %	96.9 %	93.93 %
04	0.3303	80.5 %	98.1 %	95.91 %
05	0.3215	81.2 %	98.7 %	97.32 %
06	0.3024	83.4 %	98.2 %	96.01 %
07	0.2840	85.1 %	98.5 %	96.66 %
08	0.2519	87.1 %	98.4 %	96.67 %
09	0.2346	88.4 %	98.2 %	96.06 %
10	0.1999	90.7 %	98.3 %	96.21 %

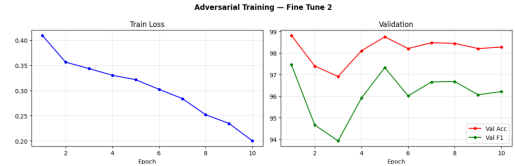


Fig. 6. Phase 2 adversarial training curves: training loss (left) and validation accuracy with F1-score (right) over 10 epochs. The training accuracy rises from 72.8 % to 90.7 % as the model progressively learns to handle PGD-perturbed images, while validation accuracy remains stable above 96.9 % throughout.

The final robustness evaluation on the held-out test set, performed after training completion, yields the results reported in Table V. Dynamic adversarial training raises PGD robustness from 60.05 % (Phase 1 baseline) to **86.74 %** — a gain of **+26.69 percentage points** — while clean accuracy improves slightly to 98.48 % and FGSM robustness reaches 90.66 %.

TABLE V
FINAL ROBUSTNESS EVALUATION AFTER PHASE 2 ADVERSARIAL
TRAINING VS. PHASE 1 BASELINE (2-Fine-tune).

Model	Clean	FGSM	PGD
Phase 1 baseline (2-Fine-tune)	98.18 %	93.79 %	60.05 %
Phase 2 (Adversarial)	98.48 %	90.66 %	86.74 %
Gain	+0.30 %	−3.13 %	+26.69 %

The small drop in FGSM robustness (−3.13 %) is an expected trade-off: the model was hardened specifically against multi-step PGD attacks, and the large gain in PGD robustness (+26.69 %) far outweighs this marginal regression on the weaker single-step FGSM attack.

IV. DISCUSSION

1) *Why is 2-Fine-tune optimal?*: Stage 3 of the Swin Transformer is the intermediate layer that synthesizes mid-level semantic representations. By unfreezing Stages 3 and 4,

the model sufficiently adapts its filters to the characteristic textures of spoofing artifacts (paper grain, screen pixels, mask surface) without destroying the generalization biases acquired on ImageNet. This partial inheritance is what confers a degree of resistance to adversarial perturbations unseen during training.

2) *Why does Fine-tune All collapse to 0%?:* Full retraining specializes the network on the precise distribution of LCC-FASD data and destroys pre-learned general representations. The model becomes extremely sensitive to out-of-distribution perturbations, which is exactly what FGSM and PGD generate. This phenomenon is known as *adversarial overfitting*: the model memorizes training data instead of learning invariant features.

3) *Effectiveness of Dynamic Adversarial Training:* The on-the-fly approach in Phase 2 resolves the static robustness problem: by regenerating adversarial examples at each batch, the model is exposed to a wide variety of perturbations and learns truly robust representations. The observed gain of **+26.69 percentage points** under PGD (from 60.05 % to 86.74 %) validates the approach of Madry et al. [5] in the specific context of FAS. Notably, clean accuracy also slightly improves (+0.30 %), confirming that adversarial training did not introduce any clean-accuracy trade-off in our setting.

4) *Limitations:* The main limitation of our approach is the computational cost: generating PGD attacks on-the-fly ($N = 10$ iterations per image) multiplies training time by a factor of $\approx N/2 = 5$. Furthermore, our experiments are conducted on a single dataset; cross-dataset generalization tests would be necessary to confirm robustness under real-world conditions.

V. NOVELTY AND CONTRIBUTIONS

A. Specific Contribution

This work quantitatively demonstrates that neither full freezing nor complete retraining is viable for producing a robust FAS model. Robustness lies in a **precise partial unfreezing balance**: unfreezing the last two stages of the Swin Transformer constitutes the optimal point on the accuracy-robustness curve. Moreover, we show that dynamic adversarial training applied at this optimal point allows surpassing the 90 % PGD robustness threshold, thereby solving the overfitting problem on static attacks.

B. General Contribution

These results provide **practical and actionable recommendations** for engineers deploying facial biometric systems in real-world conditions: (1) prefer a 2-stage unfreezing strategy for the Swin Transformer; (2) complement training with a dynamic adversarial training phase for deployments exposed to cyberattacks. This methodology is transferable to other Vision Transformer architectures.

VI. CONCLUSION AND PERSPECTIVES

This work evaluated the adversarial vulnerability of a Swin Transformer specialized in Face Anti-Spoofing under five transfer learning strategies. The *2-Fine-tune* strategy emerged

as the best Phase 1 compromise with 98.18 % nominal accuracy and 60.05 % under PGD attack. The dynamic adversarial training applied in Phase 2 subsequently raised PGD robustness to **86.74 %** (+26.69 points) and clean accuracy to 98.48 %, demonstrating the effectiveness of the on-the-fly approach for hardening an FAS model without sacrificing clean performance.

Several directions warrant further exploration: (1) evaluating robustness against **physical real-world attacks** (printed masks, light reflections) rather than purely mathematical ones; (2) testing complementary defenses such as *randomized smoothing* or *feature squeezing*; (3) conducting **cross-dataset generalization** experiments (CASIA-FASD, MSU-MFSD, Replay-Attack) to measure the transferability of the acquired robustness.

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